Income Distribution - Problem Set 1

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Use the data for the year 2018. Use data at the individual level. Answer each question for Canada and for "your group".

Ontario – Selected Group

Question 1: Variable Definitions

You will work with the following variables: total income, market income (before taxes), wage and salary income, self-employment income, age, plus the variable defining "your group". What are these variables called in the data set?

Answer:

Total income, market income (before taxes), wage and salary income, self-employment income, age, plus the variable defining "your group".

- **Total Income:** Total Income before Taxes (TTINC)
- **Market Income:** Market Income (MTINC)
- **Wage and Salary Income:** Wages and salaries before deductions (WGSAL)
- **Self-employment income:** Self-employment net income (SEMP)
- **Age:** Person's age group as of December 31 of refyear (AGEGP)
- **Reference Year (YEAR):** 2018
- **Group (Ontario):** Province (PROV) = Ontario (value = 35)

Question 2: Earnings Definition and Ratio

Consult KR's definition of earnings. Create an earnings variable in exactly the same way. (You can treat market income minus wage and salary income minus business income as "unambiguous capital income", and wage and salary income as "unambiguous labor income".) Which fraction of business income do you need to include in earnings?

Answer:

KR's definition of earnings: "We define labor earnings as wages and salaries of all kinds, plus a fraction of business income. Business income includes income from professional practices, businesses, and farm sources. The value for the fraction of business and farm income that we impute to labor earnings is the sample-wide ratio of unambiguous labor income (wages plus salaries) to the sum of unambiguous labor income and unambiguous capital income."

Variable Mapping:

- Business Income = Self-employment net income (SEMP)
- Unambiguous labour income (L_i) = Wage and salary income (WGSAL)
- Unambiguous capital income (C_i) = Market income (MTINC) – Wage and salary income (WGSAL) – Business income (SEMP)

Ratio Calculation: For each observation i, the ratio ϕ is defined as:

$$\phi = \frac{\sum_i L_i}{\sum_i (L_i + C_i)}$$

Substituting the variables:

$$\phi = \frac{\sum_i \text{WGSAL}_i}{\sum_i (\text{WGSAL}_i + (\text{MTINC}_i - \text{WGSAL}_i - \text{SEMP}_i))} = \frac{\sum_i \text{WGSAL}_i}{\sum_i (\text{MTINC}_i - \text{SEMP}_i)}$$

With Weights Included (Sample-wide): Let $w_i = \text{FWEIGHT}$.

$$\phi = \frac{\sum_i w_i \cdot \text{WGSAL}_i}{\sum_i w_i (\text{MTINC}_i - \text{SEMP}_i)}$$

Calculated Results:

- **Canada Wide ϕ:** 0.807329
- **Ontario ϕ:** 0.798375

Question 3: Replicating KR's Table 1

Table 1: Distribution of Market Income and Earnings (Canada)

Variable	Min	p1	p5	p10	p20	p40	p50	p60	p80	p90	p95	p99	Max	Mean
Market Income	-105000	0	0	0	3400	18525	28000	38250	67425	95000	121550	230500	1340000	40935.05
Earnings	-84769.51	0	0	0	0	7000	17000	29000	60000	87142.14	110000	195000	1313026	33048.04

Table 2: Distribution of Market Income and Earnings (Ontario)

Variable	Min	p1	p5	p10	p20	p40	p50	p60	p80	p90	p95	p99	Max	Mean
Market Income	-105000	0	0	0	2800	18500	28025	39000	69500	97500	125000	241875	1340000	42031.39
Earnings	-83829.34	0	0	0	0	6250	16500	29539.86	62500	87821.21	115000	197075.78	1311772.5	33556.80

Descriptive Statistics: Ontario

Replicate KR's Table 1 for market income and earnings. Also compute the mean of each. How do your two sets of results (Canada and your group) compare? How do your results compare to those for the US reported by KR?

Answer:

Comparison of Canadian and Ontario Results

The first thing to note is that the two data sets are very similar. Each metric is within only a few units (thousands) of each other.

- The mean market income and earnings for Ontario are slightly higher than Canada as a whole at \$42.03k CAD income and \$33.56k CAD earnings, compared to \$40.94k CAD and \$33.05k CAD respectively.
- Interestingly, the median market income ($p50$) is almost the exact same at \$28.00k CAD for Canada and \$28.02k CAD for Ontario.

This would suggest that the 'standard' resident in both samples has roughly the same market income. The previous two observations can largely be explained by the top-end inequality between the data sets.

- The $p99$ market income for Ontario is \$241.88k CAD while for Canada as a whole it is \$230.50k CAD.

This is the largest discrepancy within the data and shows that Ontario has a larger concentration of very high earners. One may draw the conclusion that the difference in mean income and earnings can be explained by the top-end percentiles as opposed to differences in income across the spectrum. This would also explain why the median data is so similar yet the mean data has a larger discrepancy.

Comparison of Canadian and USA Results

The first glaring difference between the data sets is the top-end of the US data.

- The US earnings data for $p90$, $p95$, $p99$ and $p100$ are \$205.7k, \$303.7k, \$969.6k and \$99,482.9k respectively.

Compared to the Canadian data these numbers are much greater. This is not unexpected, as higher levels of income inequality are well documented within the US with respect to Canada.

Another characteristic that is starkly different is the amount of 0.0 income and earnings within the Canadian data compared to the US data.

- In Canadian data: $p1$–$p20$ have 0.0 earnings and $p1$–$p10$ have 0.0 market income.
- In US data: There are no 0.0s at all.

Possible Explanations:

1. **Income Concepts:** Market income in the CIS excludes government transfers, while the data from KR measures income with a different definition.
2. **Sampling Method:** The US surveys used by KR are known to capture the very top of the income distribution more effectively (oversampling high-income households and better capturing capital income).
3. **Structural Differences:** The data sets refer to different countries, years, and currencies. Cumulative inflation in the US from 2018-2022 was approximately 16.5%. As such, each percentile in the US data would have to be reduced by 16.5% to be comparable, assuming inflation is linked with wage growth.

Question 4: Replicate KR's Figure 3b

Replicate KR's Figure 3b). Note: Here we should use total income not market income.

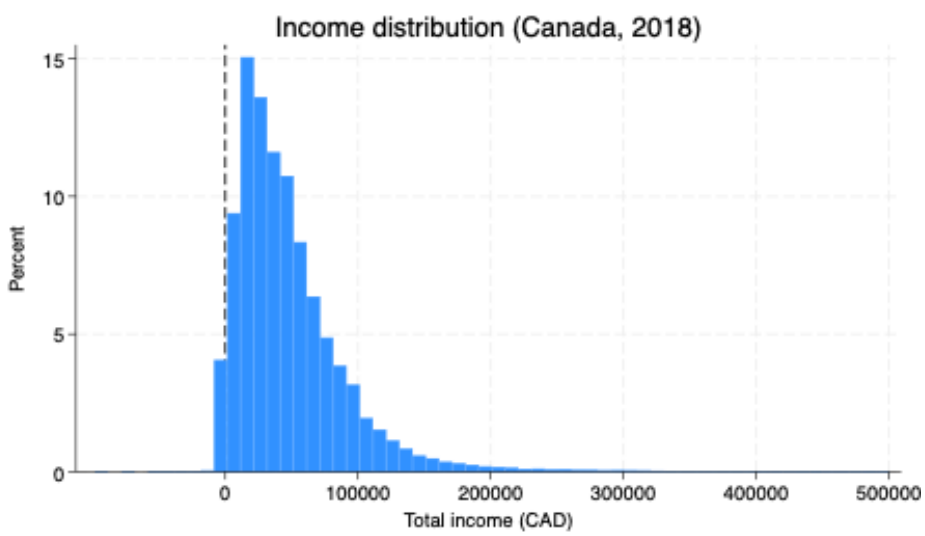

Figure 1: Income Distribution, Canada 2018

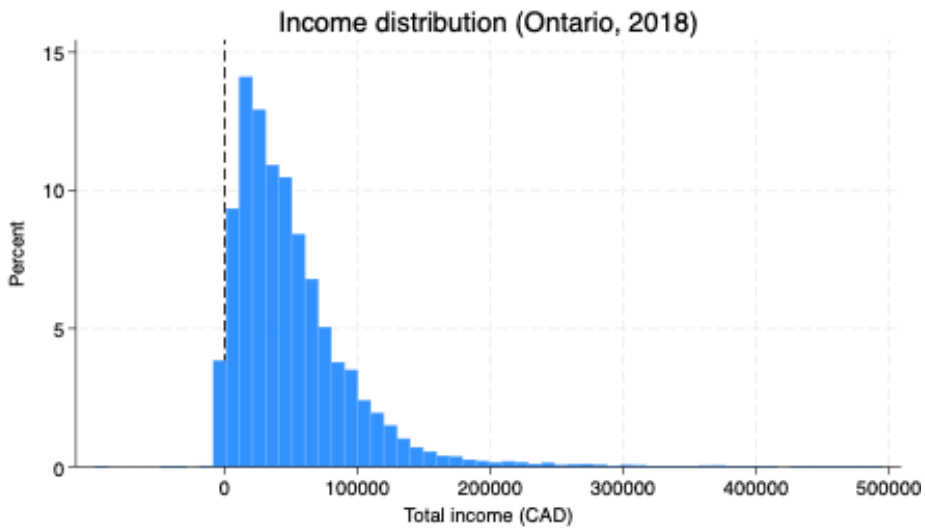

Figure 2: Income Distribution, Ontario 2018

Question 5: Inequality Metrics

Replicate KR's Table 2 for market income and earnings. How do your results compare to those for the US reported by KR?

Answer:

Table 3: Inequality Statistics: Earnings vs. Income

	Earnings	**Income**
Gini Index	0.49	0.50
Coefficient of Variation	1.38	1.20
Variance of logs	2.32	1.97
Location of the Mean (%)	50	51
Mean to Median Ratio	10.76	1.40
99–50 Ratio	5.00	7.40
90–50 Ratio	0.00	3.17
50–30 Ratio	1.88	2.61

Analysis: Canada exhibits lower inequality in both earnings and income than the US, which is to be expected.

- **Gini:** Canada Earnings 0.49 vs US 0.68; Canada Income 0.50 vs US 0.61.
- **Coefficient of Variation:** Canada Earnings 1.38 vs US 3.15; Canada Income 1.20 vs US 5.31.

- **Location of Mean:** In Canada, the mean is located around the 50th percentile for both earnings and income. In the US, it is around the 75th percentile (72nd for earnings, 78th for income).
- **Top-end Analysis:** In the US, the 99th percentile is about 17-20 times the median income/earnings. In contrast, for Canada, the 99-50 ratio is only ∼5 for earnings and ∼7.4 for income.

This huge gap indicates that the very rich in the US are much further above the median household than their Canadian counterparts.

Question 6: Income Shares

Replicate the table in the slide entitled "Sources of income across the distribution" from Lecture 2, sorting persons by market income. Report income shares of wage and salary income and business income. How do your two sets of results (Canada and your group) compare?

Answer:

Table 4: Income Shares from Wages and Business Income by Income Group (Canada vs Ontario)

Income Group	**Wage Share**		**Business Share**	
	Canada	**Ontario**	**Canada**	**Ontario**
Q1 (bottom 20%)	0.66	0.66	-0.56	-0.49
Q2	0.52	0.54	0.09	0.08
Q3	0.64	0.66	0.05	0.05
Q4	0.74	0.77	0.03	0.03
Q5 (80–95%)	0.84	0.86	0.03	0.03
Top 2–5%	0.76	0.80	0.06	0.05
Top 1%	0.69	0.72	0.07	0.07

Analysis: Overall, the sets are quite similar with some slight differences.

- Wage income dominates for the majority of the distribution. Wage share increases with income (for the most part) from around 0.52–0.54 in Q2 to 0.84–0.86 in Q5.
- In Ontario, wage shares are slightly higher across most groups, indicating a slightly more wage-dominated income structure compared to the national average.
- Business income plays a small but growing share in the upper tail, reaching 0.07 for the top 1% in both samples.

- In Q1, the business share is negative (-0.56 for Canada and -0.49 for Ontario), likely reflecting net business losses.
- Wage share declines slightly at the very top of the distribution. It peaks at 0.84–0.86 in the 80–95% group and drops to 0.69 for Canada and 0.72 for Ontario. The gap between the rest of the income (e.g., $0.07 + 0.69 \neq 1.0$) can probably be explained by capital income; the highest earners could easily be making ∼20% of their overall income from capital.

Question 7

Replicate the figures "The role of age" and "Inequality across cohorts (Gini)" from Lecture 2, for earnings.

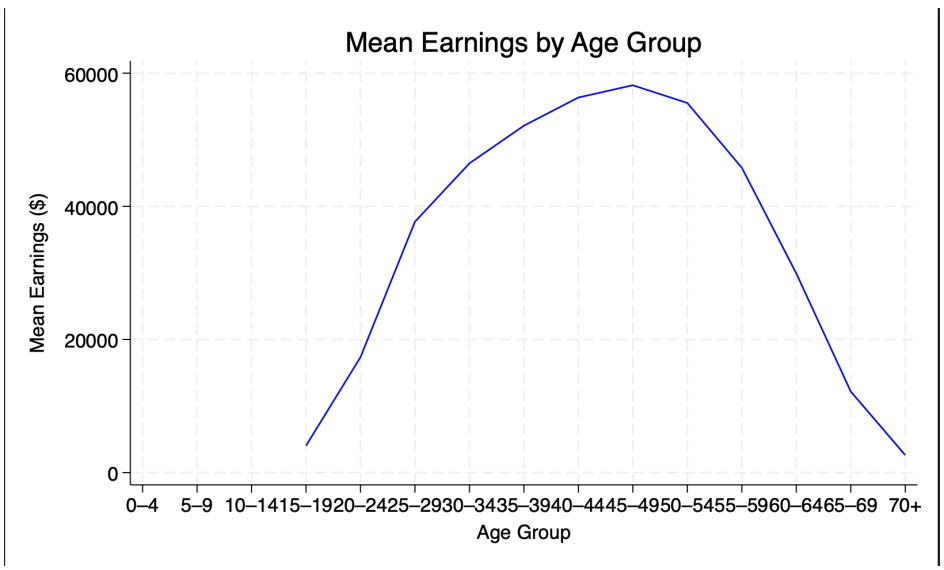

Figure 3: The Role of Age

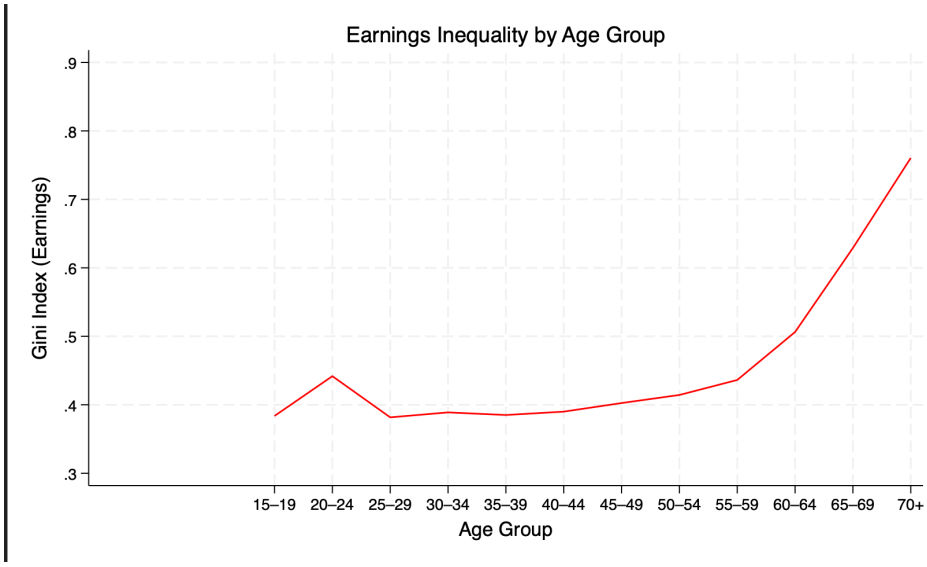

Figure 4: Inequality across cohorts